

34.4 Iteratively reweighted least squares

The likelihood equations Eq. 34.3.1 have no closed-form solution. Any general-purpose optimizer will solve them, but there is a better way, and it is why generalized linear models became practical: the natural Newton-type iteration is a *weighted least squares fit*, repeated. The apparatus of Chapter 6 and Section 21.3 is therefore reusable, including the stable routines of Chapter 10.

34.4.1 Fisher scoring is weighted least squares

Newton's method for maximizing ℓ replaces the observed information by its inverse: $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \{-\nabla^2 \ell\}^{-1} \mathbf{U}$. **Fisher scoring** replaces the observed information by the expected information $\mathcal{I}$ instead. By Theorem 34.3.1(d) the two are the same under the canonical link; in general they differ, and the expected version is both simpler and guaranteed nonnegative definite.

Theorem 34.4.1. Fisher scoring is iteratively reweighted least squares

Let $\boldsymbol{\beta}^{(t)}$ be in the interior of $\mathcal{B}$ with $\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}$ nonsingular, where $\mathbf{W}^{(t)}$ is the working-weight matrix of Theorem 34.3.1 evaluated at $\boldsymbol{\beta}^{(t)}$. Define the **working response**

$$z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \mu_i^{(t)}}{h'(\eta_i^{(t)})} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) g'(\mu_i^{(t)}). \quad (34.4.1)$$

Then the Fisher scoring step $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \mathcal{I}(\boldsymbol{\beta}^{(t)})^{-1} \mathbf{U}(\boldsymbol{\beta}^{(t)})$ is exactly the weighted least squares estimate

$$\boldsymbol{\beta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{z}^{(t)}, \quad (34.4.2)$$

that is, the regression of the working response $\mathbf{z}^{(t)}$ on $\mathbf{X}$ with weights $\mathbf{W}^{(t)}$. The dispersion ϕ cancels and never enters the iteration.

Proof. By Eq. 34.3.2 and Eq. 34.3.1, with everything evaluated at $\boldsymbol{\beta}^{(t)}$,

$$\begin{aligned} \mathcal{I}^{-1} \mathbf{U} &= \phi (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \cdot \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}) \\ &= (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}), \end{aligned}$$

so ϕ has already cancelled. Adding $\boldsymbol{\beta}^{(t)} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X} \boldsymbol{\beta}^{(t)}$ and using $\mathbf{X} \boldsymbol{\beta}^{(t)} = \boldsymbol{\eta}^{(t)}$,

$$\boldsymbol{\beta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \{\boldsymbol{\eta}^{(t)} + \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu})\},$$

and the braced vector is $\mathbf{z}^{(t)}$ by Eq. 34.4.1, since $\mathbf{D}^{-1} = \text{diag}\{1/h'(\eta_i)\}$ and $g'(\mu) = 1/h'(\eta)$. ■

Idea. The working response is a first-order Taylor expansion of the link applied to the data: $g(y_i) \approx g(\mu_i) + (y_i - \mu_i)g'(\mu_i) = z_i$. So each step regresses “the data, mapped to the linear-predictor scale” on $\mathbf{X}$. The working weight is the reciprocal of that quantity’s approximate variance, because

$$\text{Var}(z_i) \approx g'(\mu_i)^2 \text{Var}(Y_i) = \frac{\phi}{w_i} \frac{V(\mu_i)}{h'(\eta_i)^2} = \frac{\phi}{\mathbf{W}_{ii}}.$$

Both the response and the weights depend on the current fit, which is why the regression has to be iterated.

Written out, the algorithm is short.

Algorithm.

Iteratively reweighted least squares.

1. Choose starting fitted means $\mu_i^{(0)}$ and set $\eta_i^{(0)} = g(\mu_i^{(0)})$.
2. For $t = 0, 1, 2, \dots$: compute $\mathbf{W}_{ii} = w_i h'(\eta_i^{(t)})^2 / V(\mu_i^{(t)})$ and $z_i^{(t)}$ from [Eq. 34.4.1](#); solve the weighted least squares problem [Eq. 34.4.2](#) for $\boldsymbol{\beta}^{(t+1)}$; set $\boldsymbol{\eta}^{(t+1)} = \mathbf{X}\boldsymbol{\beta}^{(t+1)}$ and $\mu_i^{(t+1)} = h(\eta_i^{(t+1)})$.
3. Stop when $\boldsymbol{\beta}$ (or the deviance of [Section 34.5](#)) stops changing; report $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}^{(t+1)}$ and $(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$, with $\hat{\mathbf{W}}$ the weights at the solution.

Because the map is a fixed-point iteration whose fixed points solve [Eq. 34.3.1](#), any limit is a solution of the likelihood equations; under the canonical link with $\text{rank}(\mathbf{X}) = p$, [Theorem 34.3.3](#) then makes it *the* maximum likelihood estimate.

Starting values. The usual device is to start from the data themselves: set $\mu_i^{(0)} = y_i$, adjusted if necessary to keep it inside $\mathcal{M}$. For a Poisson response take $\mu_i^{(0)} = y_i + 0.1$; for a binomial proportion, $\mu_i^{(0)} = (w_i y_i + \frac{1}{2}) / (w_i + 1)$, the empirical-logit correction of Exercise (B2); for a gamma response, $\mu_i^{(0)} = y_i$. The first step is then the one-step weighted least squares fit that preceded generalized linear models historically, and the iteration refines it.

Relation to the least squares chapters. Each step of [Eq. 34.4.2](#) is the weighted least squares problem of [Theorem 21.3.1](#) with *known* weights, so its convergence theory is numerical, not statistical: the final $(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$ is the inverse expected information of the likelihood, not the covariance of a fixed-weight estimate, and the two coincide asymptotically for a different reason ([Theorem 34.5.1](#)). Compute the step by the QR or Cholesky routines of [Section 6.10](#) applied to $\mathbf{W}^{1/2} \mathbf{X}$ and $\mathbf{W}^{1/2} \mathbf{z}$, never by inverting $\mathbf{X}^\top \mathbf{W} \mathbf{X}$.

Warning. The name “iteratively reweighted least squares” is also used for the robust M-estimation algorithm of [Proposition 20.6.1](#), and the two are different. There the weights come from a ρ function chosen to resist outliers, and the iteration provably decreases the objective at every step. Here the weights come from the variance function, the iteration is Fisher scoring, and there is no monotonicity guarantee unless the link is canonical.

34.4.2 The algorithm in practice

The listing below implements [Eq. 34.4.2](#) in a dozen lines. A family is nothing but (V, g, h) together with h' and a starting rule; the solver never needs to know which distribution it is fitting, which is the computational content of the whole chapter.

```
import numpy as np
from scipy import stats

# A family is the variance function V, the link g, its inverse h, the derivative
# h' = d(mu)/d(eta), and a rule for the starting fitted means.
def poisson_log():
    return dict(name="Poisson/log", V=lambda mu: mu, g=np.log, h=np.exp,
               dmu=lambda eta: np.exp(eta), start=lambda y, w: y + 0.1)

def binomial_logit():
    return dict(name="binomial/logit", V=lambda mu: mu * (1 - mu),
               g=lambda mu: np.log(mu / (1 - mu)), h=lambda eta: 1 / (1 + np.exp(-
eta))),
               dmu=lambda eta: 1 / (2 + np.exp(eta) + np.exp(-eta)),
               start=lambda y, w: (w * y + 0.5) / (w + 1.0))

def binomial_probit():
    return dict(name="binomial/probit", V=lambda mu: mu * (1 - mu),
               g=stats.norm.ppf, h=stats.norm.cdf, dmu=stats.norm.pdf,
               start=lambda y, w: (w * y + 0.5) / (w + 1.0))

def gamma_log():
    return dict(name="gamma/log", V=lambda mu: mu ** 2, g=np.log, h=np.exp,
               dmu=lambda eta: np.exp(eta), start=lambda y, w: y)

def irls(X, y, family, w=None, tol=1e-12, max_iter=60, trace=False):
    """Fisher scoring for a generalized linear model, as weighted least squares.

    At each step the working response is  $z = \eta + (y - \mu)/h'(\eta)$  and the working
    weight is  $W = w h'(\eta)^2 / V(\mu)$ ; beta is the weighted least squares
    coefficient
    of z on X. The dispersion phi never enters, because it cancels.
    """
    w = np.ones(len(y)) if w is None else np.asarray(w, float)
    mu = family["start"](y, w)
    eta = family["g"](mu)
    beta, path = np.zeros(X.shape[1]), []
    for _ in range(max_iter):
        d = family["dmu"](eta)
        W = w * d ** 2 / family["V"](mu)
        z = eta + (y - mu) / d
        XtW = X.T * W
        beta_new = np.linalg.solve(XtW @ X, XtW @ z)
        path.append(beta_new)
        eta = X @ beta_new
        mu = family["h"](eta)
        if np.max(np.abs(beta_new - beta)) < tol * (1 + np.max(np.abs(beta_new))):
            beta = beta_new
            break
        beta = beta_new
    W = w * family["dmu"](eta) ** 2 / family["V"](mu)
```

```
cov_unscaled = np.linalg.inv((X.T * W) @ X) #  $(X^T W X)^{-1}$ 
return (beta, cov_unscaled, np.array(path)) if trace else (beta, cov_unscaled)
```

[Run this code online](#)

```
import statsmodels.api as sm

rng = np.random.default_rng(3401)
n = 300
Xs = np.column_stack([np.ones(n), rng.normal(size=n), rng.binomial(1, 0.4, n)])
beta_true = np.array([0.7, 0.5, -0.4])
counts = rng.poisson(np.exp(Xs @ beta_true))

beta_hat, cov = irls(Xs, counts.astype(float), poisson_log())
sm_fit = sm.GLM(counts, Xs, family=sm.families.Poisson()).fit()
print("from scratch:", np.round(beta_hat, 6))
print("statsmodels :", np.round(sm_fit.params, 6))
print("standard errors:", np.round(np.sqrt(np.diag(cov)), 6), np.round(sm_fit.bse,
6))
```

[Run this code online](#)

The script behind this section runs the same solver on four models — Poisson/log, binomial/logit, binomial/probit and gamma/log — and checks the coefficients and standard errors against `statsmodels` to at least eight decimal places in every case. Two details matter. For the binomial fits the response passed to the solver is the *proportion* s_i/m_i with prior weight $w_i = m_i$, which is Example (The binomial as a family of proportions) in code. For the gamma fit the solver returns $(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$, so the standard errors need the extra factor $\hat{\phi}$, estimated separately; on Engel's data that gives $\hat{\boldsymbol{\beta}} = (0.5068, 0.8629)$ with $\hat{\phi} = 0.0177$ and a standard error of 0.0198 for the elasticity.

34.4.3 Convergence and its failures

Example. Two convergence rates

Figure 34.4.1 records $\max_j |\beta_j^{(t)} - \hat{\beta}_j|$ against the iteration count for the four models. The canonical-link fits reach full double precision in 5 and 4 steps, with error curves that bend downwards on the logarithmic scale: the correct digits roughly double at each step. The non-canonical fits take 5 and 7 steps, along straight lines, so their correct digits grow by a fixed amount per step.

This is what the theory predicts. Under the canonical link Fisher scoring *is* Newton's method (Theorem 34.3.1(d)), quadratically convergent near a nondegenerate maximum. Under any other link the two information matrices differ by $\phi^{-1} \mathbf{X}^\top \mathbf{D}_r \mathbf{X}$, which vanishes only in expectation, so the iteration is Newton's method with a perturbed Hessian and converges linearly — in practice, one or two extra iterations.

Four things go wrong often enough to be worth recognizing.

The estimate does not exist. When the maximum of Example (Two ways the maximum fails to exist) is not attained the iteration cannot converge, and its behaviour is characteristic: the coefficients grow without limit, the deviance falls towards its infimum, the working weights

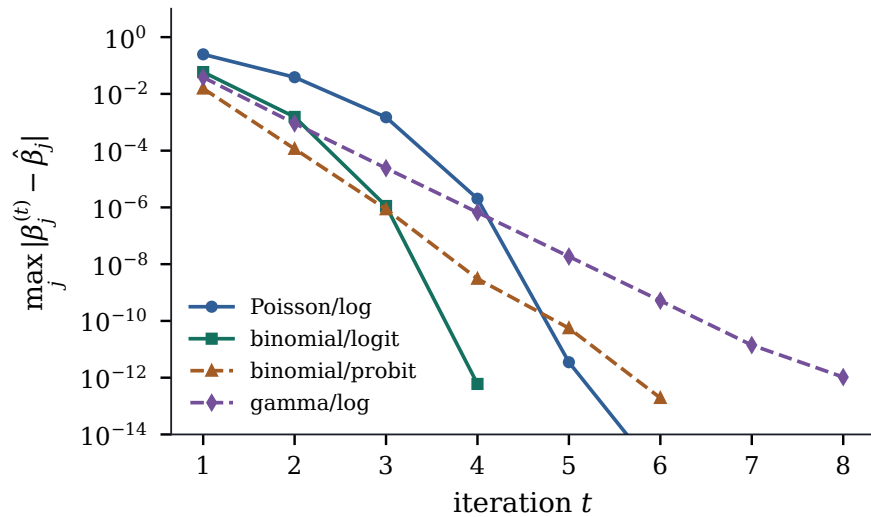


Figure 34.4.1: Convergence of iteratively reweighted least squares for four models. Solid curves are canonical links, dashed curves are not. On a logarithmic scale a quadratic rate bends downwards and a linear rate is straight.

collapse to zero. Software that stops on a small change in the deviance then reports a large but finite estimate with a large standard error and no warning.

Example. What separation looks like from inside the algorithm

Ten binary responses were generated with $y_i = 1$ exactly when $x_i > 0$, so the data are completely separated. Running the solver on them gives

step	$\hat{\beta}_1$	deviance	largest working weight
1	1.36	4.589	0.245
4	5.74	0.931	0.159
7	16.95	0.0574	0.0142
8	—	—	—

At the eighth step every fitted probability has reached 0 or 1 in double precision, so $V(\hat{\mu}_i) = 0$, the working weights are 0/0, and the iteration returns nothing. The slope was never going to settle: the supremum of the likelihood is approached only as $\beta_1 \rightarrow \infty$. [Chapter 35](#) characterizes exactly when this happens and what to do about it (Theorem 35.3.1).

```
x_sep = np.array([-2.4, -1.7, -1.1, -0.6, -0.2, 0.3, 0.8, 1.4, 1.9, 2.6])
y_sep = (x_sep > 0).astype(float) # the two groups do not
                                overlap
X_sep = np.column_stack([np.ones(10), x_sep])

for iters in range(1, 9):
    with np.errstate(divide="ignore", invalid="ignore"):
        b, _ = irls(X_sep, y_sep, binomial_logit(), tol=0.0, max_iter=iters)
        mu = 1 / (1 + np.exp(-(X_sep @ b)))
        dev = -2 * np.sum(np.where(y_sep == 1, np.log(mu), np.log(1 - mu)))
    print(f"{iters} steps: slope {b[1]:9.4f}    deviance {dev:.3e}"
          f"    largest working weight {np.max(mu * (1 - mu)):.3e}")
```

[Run this code online](#)

The step leaves the admissible set. With a link whose range is not all of $\mathbb{R}$, a full step can produce $\eta_i^{(t+1)} \notin g(\mathcal{M})$ — a negative fitted mean for an identity-link Poisson model, where the variance function is then negative and the “weights” meaningless. The repair is **step halving**: if the proposed $\boldsymbol{\beta}^{(t+1)}$ is inadmissible, or the deviance increases, halve the step and retry. A few halvings almost always suffice, and software applies them automatically.

The information matrix is nearly singular. The matrix inverted at each step is $\mathbf{X}^\top \mathbf{W} \mathbf{X}$, not $\mathbf{X}^\top \mathbf{X}$, so full column rank is no longer enough: if the weights are tiny wherever a regressor varies, the weighted matrix can be numerically singular even for a well-conditioned $\mathbf{X}$. The diagnostics of [Chapter 26](#) apply to $\mathbf{W}^{1/2} \mathbf{X}$, not $\mathbf{X}$.

The iteration converges to the wrong stationary point. For non-canonical links the log-likelihood need not be concave, and Fisher scoring from a bad start can settle at a local maximum; the defence is several starting values compared by maximized log-likelihood.

34.4.4 Exercises

34.4.4.1 A. Check your understanding

Exercise. A1

Verify the approximation $\text{Var}(z_i) \approx \phi/\mathbf{W}_{ii}$ stated in the key-idea box, and explain which two approximations are being made.

34.4.4.2 B. Practice

Exercise. B1

Take grouped binomial data with all $0 < s_i < m_i$ and start the iteration at $\mu_i^{(0)} = s_i/m_i$. Show that the first working response is the empirical logit $\log\{s_i/(m_i - s_i)\}$ and the first working weight is $s_i(m_i - s_i)/m_i$. Compare with Exercise (B2).

Solution. With $\mu_i^{(0)} = s_i/m_i$ the starting linear predictor is already $\eta_i^{(0)} = \log\{\mu_i^{(0)}/(1 - \mu_i^{(0)})\} = \log\{s_i/(m_i - s_i)\}$, and $y_i - \mu_i^{(0)} = 0$, so $z_i^{(0)} = \eta_i^{(0)}$ is the empirical logit. The working weight is $\mathbf{W}_{ii} = m_i \mu_i^{(0)} (1 - \mu_i^{(0)}) = s_i(m_i - s_i)/m_i$, which is the weight of Exercise (B2) with $\hat{p}_i = s_i/m_i$. The first iterate is therefore exactly the empirical-logit weighted least squares estimate; later iterates are not, which is why the two methods give different answers in small samples.

Exercise. B2

Modify [Eq. 34.4.2](#) for a model with an offset $\eta_i = o_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, as in Exercise (C2). Show that the working response becomes $z_i = \eta_i - o_i + (y_i - \mu_i)g'(\mu_i)$ and nothing else changes.

34.4.4.3 C. Going deeper

Exercise. C1

Show that under the canonical link, Fisher scoring is Newton's method applied to a strictly concave function, and deduce that when the maximum exists, the iteration converges to it from any starting point at which the step is well defined *provided the step is damped*: for each t there is $s_t \in (0, 1]$ with $\ell(\boldsymbol{\beta}^{(t)} + s_t \mathbf{d}^{(t)}) > \ell(\boldsymbol{\beta}^{(t)})$, where $\mathbf{d}^{(t)}$ is the Newton direction. Why does the undamped step not carry the same guarantee? (Compare Proposition 20.6.1, where the undamped step *is* monotone, because the argument there is majorization rather than Newton.)

Solution. By Theorem 34.3.3 the Hessian $-\mathcal{I}$ is negative definite, so the Newton direction $\mathbf{d} = \mathcal{I}^{-1}\mathbf{U}$ satisfies $\mathbf{U}^\top \mathbf{d} = \mathbf{U}^\top \mathcal{I}^{-1}\mathbf{U} > 0$ whenever $\mathbf{U} \neq \mathbf{0}$: it is an ascent direction, so the directional derivative of ℓ at $\boldsymbol{\beta}^{(t)}$ along $\mathbf{d}$ is positive and some small enough step increases ℓ . The undamped step has length fixed by the local quadratic approximation, which can overshoot badly when ℓ is far from quadratic, so it may decrease ℓ or leave $\mathcal{B}$ altogether.